

Agent Memory

that works with

your existing Search Stack

Haystack EU · Berlin, Main Stage · Wed 16 September 2026, 16:00

Paul-Louis Nech · Algolia

At least I remembered the interleaving tip.

Remember everything?

MEMORY

User prefers Vue.js over React
for new repos

technical

2026-08-12



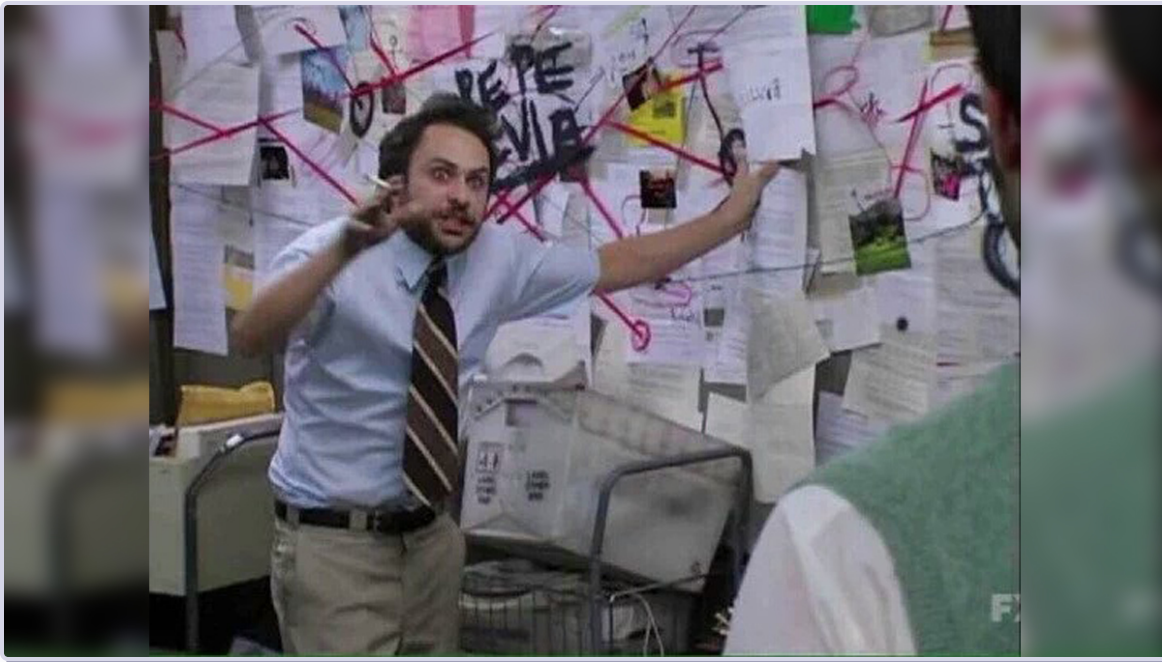
INTRO



This year I briefed my agents. Every. Single. Conversation.

Remembering not enough is bad.

The opposite is true too: bringing everything up is bad as well.



A CHATGPT SYSTEM PROMPT

```
[system]
...
model set context:
  2025-04-11: user prefers concise answers...
  2025-04-12: user is building a Vue app...
  2025-04-14: user lives in Paris...
  ... 40+ lines, every single turn ...
```

Remembering nothing is bad.

**Remembering everything
is bad too.**

False positives.

False negatives.

It's an F1 score. ✨

Memory is a Retrieval Problem

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BTW hi 🖐️

Paul-Louis Nech · Staff ML Engineer, Algolia @

We do Search.

(and Recommendations)

(and more)

Agent Studio is the
agentic harness on top.



OUR TURF

Our turf

Our turf

9 min

Components

12 min

Side by side

5 min

Conclusion

7 min

**Agents are new.
Retrieval under constraints
is **our** turf.**

Budget 1: latency



Budget 2: tokens

Token bill · estimated		
MEMORIES IN THE PROMPT	≈ TOKENS	× A 20-TURN CHAT
4	550	11k
10	1.4k	28k
30	4.2k	83k

You pay for every over-fetched memory, on every single turn.



Budget 3: attention

Our needles · 2025 vs 2026		
	2025	2026
needle lost at	20k	never
128k costs you	a wrong answer	75–134 s

In 2025 it was impossible.
Now it's just stupid expensive.

Where benchmarks give you a baseline

thus a first impression



LoCoMo arXiv 2402.17753



LongMemEval arXiv 2410.10813

One answer, four turns, three sessions

ONE QUESTION, VERBATIM

Q

What activities does Melanie partake in?

A

pottery, camping, painting, swimming

ev

D1:12

D1:18

D5:4

D9:1

No string match scores that answer.



LoCoMo

arXiv 2402.17753

LoCoMo, as released	
conversations	10
sessions each	19–32
questions	1,986

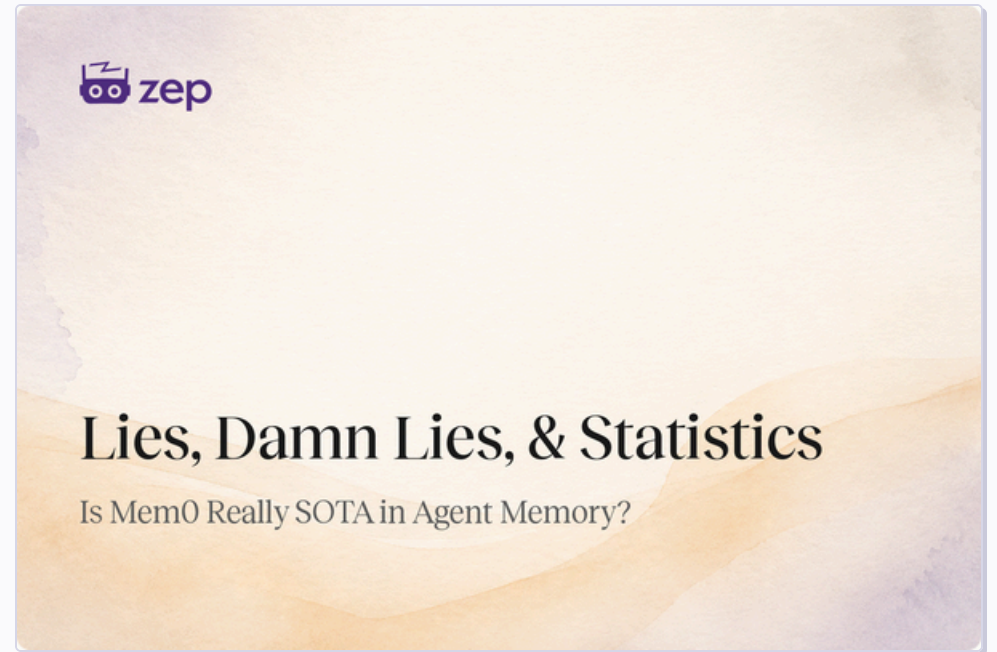
Benchmarks tell a partial picture

What the score hides

ten conversations, 25k tokens at most

one question in five has no answer

full context beat the memory systems



first two measured in locomo10.json on 09-16 · third reported by Zep, May 2025

OUR TURF

What LongMemEval asks

What it asks · n=500	
multi-session	133
temporal-reasoning	133
knowledge-update	78
single-session-user	70
single-session-assistant	56
single-session-preference	30

ONE QUESTION, VERBATIM

Q What degree did I graduate with?
A Business Administration

One of 54 sessions holds it.

AND ONE IT MUST REFUSE

Q What is the name of my hamster?
A You did not mention
 this information.

You mentioned your cat Luna but not
your hamster.

COMPONENTS

“Lies, Damned Lies, and Benchmarks”

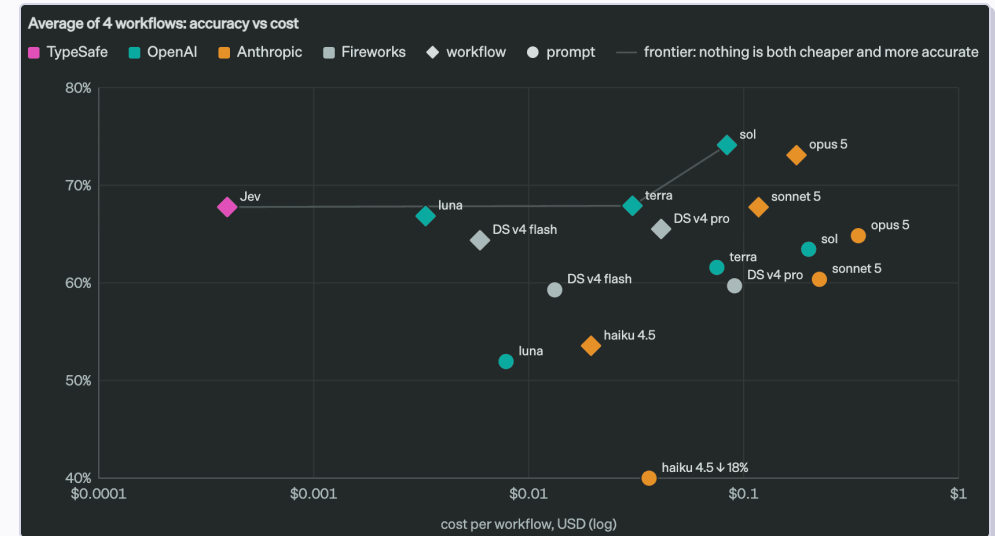
Benchmark Lies

Models chase benchmarks.

Benchmarks chase users.

People cherry-pick for their favorite narrative.

Benchmarks have been broken for a while.



“De-emphasizing benchmarks
even when you're ahead.”

typesafe.ai/blog/antibenchmaxxing · their words and their chart, on their own model release

COMPONENTS

...and where your shopper doesn't care.

and has different success metrics



You already have query → results.

Components

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① The query.



four ways to build it

verbatim?

with topics?

saved triggers?

LLM rewrite?

COMPONENTS



Baseline: just the last N.



One conversation. Five queries.

TIRED	the last N memories	– no query at all
WIRED	the user, verbatim	"what haven't i tasted yet?"
+	widen with topics	... · topics: food, preferences
+	phrases saved at write time	recallTriggers: "coffee not tried yet"
HIRED	LLM rewrite, one round-trip	"specialty coffee Paris untried"

Same ask. The query is a design decision, not an inheritance.

The query you wrote at write time.

memory "... been to every specialty coffee bar on their Paris list
 except Substance in the 3rd ..."

triggers "what coffee haven't i tried yet"
 "specialty coffee, 3rd arrondissement"
 "places still on my list"

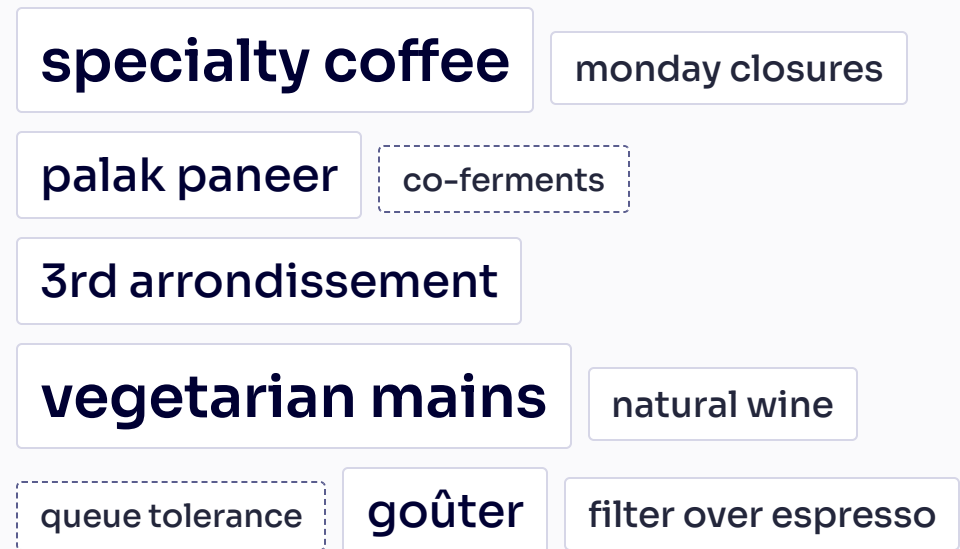
Triggers: simplified example.

Write the query when you write the memory.

...especially with topics.



closed: our real 18, one set, filterable



open: the corpus writes its own

COMPONENTS

② Retrieval.



levers you already tune

ranking formula

facets & filters

optional scoring

rules · synonyms

typo tolerance

📖 Docs

COMPONENTS

Same words: fine.
Same meaning: not so much.

Recall@1 · n=625/644/644	
the memory's own words	81%
rephrased	34%
intent, no shared words	9.5%
same index. same records.	

Three settings that moved recall.

Twelve swept · n=1913	
SETTING	Δ RECALL@10
removeStopWords=false	-35.4 pp
searchableAttributes=[text]	-5.4 pp
optionalWords dropped	+2.9 pp
the other nine	noise

③ Formatting.



what you hand the model

attributesToRetrieve

snippeting

record format?

SIDE BY SIDE

JSON? TOON? YAML? Markdown?

JSON

```
{"hits": [{"objectID": "d0cd1026e8ae51ef342eddaa108faeed2a1f6e2d0386dd1756a75e7fcfba34e2", "text": "User completed their undergraduate degree in Computer Science from UCLA.", "rawExtract": "I completed my undergrad in CS from UCLA, which has a great reputation in the industry.", "memoryType": "semantic", "createdAt": "1788961514", "keywords": "UCLA, Computer Science, undergraduate, bachelor degree..."}]}
```

TOON

```
hits[4]{objectID,text,rawExtract,memoryType,createdAt,keywords,topics,recallTriggers}:  
d0cd1026e8ae51ef342eddaa108faeed2a1f6e2d0386dd1756a75e7fcfba34e2,User completed their undergraduate degree in Computer Science from UCLA., "I co..."
```

same 4 memories, ~11% fewer tokens

Measure it.

Answer accuracy % · n=60 per arm	
json	53.3
toon	58.3 (+5.0)
yaml	56.7
markdown_kv	56.7
not significant	

Query · Retrieval · Formatting.



Side by side

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CONCLUSION



Same question. Three agents.

Four seasons of Memory

our own stage rig · one question · September 2026 · tokens exact, clocks indicative			
	SECONDS	TOKENS IN	FACTS RECALLED
no memory	2.9	65	0
the last 5 turns	5.4	345	1
memory as a tool	18.1	9,184	1
preload + preflight	7.5	345	3

Tools-only: 27× tokens, yet less context.

Conclusion

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CONCLUSION

**Some players go pure vector.
Sub-200 ms. Impressive!**

**First: make it work on
what you **already** have.**

Forgetting is retrieval too.

I asked the agent to compact its own memory.

91% of deletions destroyed unique anchors.



Your Turn: Measure It

Read 5 conversations.
Read 20 memories.
Search twice yourself.

Retrieved, irrelevant?

Relevant, not retrieved?

**Memory is Retrieval,
and retrieval is our domain.**

The screenshot displays a search evaluation interface. The top section, titled 'RETRIEVED RECORDS', lists 7 records with their IDs and associated keywords. The bottom section, titled 'JUDGMENTS FOR THIS CASE', shows a score of 0.77, a 'PASSED' status, and a 'RELEVANCE v2' rubric. It also includes a 'REASON (VERBATIM)' section with a redacted area, and a 'RETRIEVAL RECALL' section showing a score of 1.00. The 'EXPECTED VS GOT' section compares expected and actual results for various keywords, with some marked as correct (green checkmarks) and others as incorrect (red X marks). The 'JUDGMENT METADATA' section shows the final score calculation: $\sqrt{(1.00 \times 0.60)} = 0.77$.

Record ID	Keywords
17700562-8481	neuralsearch, vector, keyword
15032552-8161	neuralsearch, keyword, documentation
en_253cf_0_2	keyword
en_90460_0_1	keyword, website
en_81183_at_1	neuralsearch, vector, keyword, website
en_7cf43_0_1	no target
en_1056a_0_2	keyword

JUDGMENTS FOR THIS CASE · 1

0.77 **PASSED** **relevance v2** **RUBRIC v5** **JUDGE deterministic** **SOURCE deterministic** 12d ago

REASON (VERBATIM)

retrieval_recall=1.00 response_recall=0.60

RETRIEVAL RECALL **1.00** **RESPONSE RECALL** **0.60** axes

EXPECTED VS GOT derived in console

EXPECTED IN RETRIEVAL neuralsearch ✓ vector ✓ keyword ✓ website ✓ documentation ✓

EXPECTED IN ANSWER neuralsearch ✓ vector ✓ keyword ✓ website ✗ documentation ✗

retrieval 5/5 · response 3/5 · score = $\sqrt{(1.00 \times 0.60)} = 0.77$

JUDGMENT METADATA

(payload)

Memory is just search.

THE QUESTION THE AGENT WAS ASKED

What is NeuralSearch?

019fc709..d70d

DIFFICULTY easy

LANG en

APPROVED

ANSWER KEY (CASEPAYLOAD.EXPECTED)

SHOULD CONTAIN CATEGORIES

Website

Documentation

KEYWORDS IN RESULTS

NeuralSearch

vector

keyword

(case payload)

RECEIPT · EXECUTION

MESSAGE 01a04946...e835

alg_msg_54ee15TCneElfXbv

CONVERSATION 01a04946...fc3d

EXECUTED 2026-08-28 16:49:28

(payload)

RECEIPT · JUDGING

JUDGMENT 01a04946...fd80

relevance

0.77

PASSED

JUDGED 2026-08-28 16:49:28

(payload)

WHAT THE AGENT DID · STORED COMPLETION

RESPONSE TEXT

NeuralSearch is Algolia's AI-powered hybrid search approach. It combines:

- **Keyword matching** for precise terms and exact wording

- **Semantic matching** to understand the concept or intent behind a query

- **Neural ranking** that combines keyword and semantic scores to determine relevance

It uses vectors and can learn from your own click and conversion events, helping results improve based on how users interact with your content. It's especially useful when users search naturally or use terms that don't exactly appear in your records.

NeuralSearch is less suited to use cases based primarily on exact matches, such as SKU-only searches.

Are you evaluating NeuralSearch for ecommerce, content, or another search experience?

TOOL CALLS · 1

algolia_search_index_algolia_www_prod_v2

arguments

RETRIEVED RECORDS · 7 · badges = the expected targets this record's own text contains

1 How does NeuralSearch rank results?

17700562..8481

neuralsearch

vector

keyword

Paul-Louis Nech · Agentic Memory on your existing Search Stack · Haystack 26 · 43/44

Thanks.

✉ PLN@algolia.com

🐙 [@PLNech](https://github.com/PLNech)

★ [Check our Agent Studio](#) · [Straight to the docs](#)



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